SARVAA Fresh
Database Architecture Document (DATABASE.md)
Version: 1.0
Database: PostgreSQL 16+
ORM: Prisma ORM
ID Strategy: UUID
Soft Delete Strategy: Enabled

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1. Database Design Principles
Core Principles
Category Agnostic
Products must support:

- **Fruits**
- **Vegetables**
- **Dairy**
- **Grocery**
- **Household**
- **Future Categories**

without code changes.

Variant Based Inventory
Inventory belongs to product variants.
Example:
Tomato

- **500g**
- **1kg**
- **2kg**

Each variant has separate stock.

Event-Based Order Tracking
Order history must be preserved.
Example:
Placed
↓
Confirmed
↓
Packed
↓
Out For Delivery
↓
Delivered

Auditability
Every table contains:
created_at
updated_at
Most business tables contain:
deleted_at
for soft delete.

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2. Core Entity Relationship Overview

```
users
 |
 |---- addresses
 |
 |---- orders
 |
 |---- reviews (future)
 |
 |---- loyalty_transactions (future)

categories
 |
 |---- products
       |
       |---- product_variants
             |
             |---- inventory
             |
             |---- order_items

orders
 |
 |---- order_items
 |
 |---- payments
 |
 |---- order_status_history

admins
 |
 |---- inventory_updates
 |
 |---- order_updates
```

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3. USERS

Purpose:

Customer Accounts

Fields

Field	Type
id	UUID PK
full_name	VARCHAR(150)
mobile	VARCHAR(15) UNIQUE
email	VARCHAR(255) NULL
password_hash	TEXT
is_active	BOOLEAN

Field	Type
created_at	TIMESTAMP
updated_at	TIMESTAMP
deleted_at	TIMESTAMP NULL

Indexes
mobile UNIQUE
email
Relationship:
User
├── Addresses
├── Orders
└── Future Reviews

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4. ADDRESSES
Purpose:
Multiple customer delivery addresses
Fields

Field	Type
id	UUID PK
user_id	UUID FK
label	VARCHAR(50)
full_address	TEXT
city	VARCHAR(100)
state	VARCHAR(100)
pincode	VARCHAR(10)
landmark	VARCHAR(255)
is_default	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP

Foreign Keys
user_id → users.id

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5. CATEGORIES
Purpose:
Product Classification
Examples:

- **Fruits**
- **Vegetables**
- **Dairy**
- **Grocery**

Fields

Field	Type
id	UUID PK
name	VARCHAR(100)
slug	VARCHAR(120) UNIQUE

Field	Type
description	TEXT
image_url	TEXT
is_active	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP

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6. PRODUCTS
Purpose:
Master Product Information
Example:
Tomato
Milk
Rice
Fields

Field	Type
id	UUID PK
category_id	UUID FK
name	VARCHAR(255)
slug	VARCHAR(255) UNIQUE
description	TEXT
image_url	TEXT
is_active	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP
deleted_at	TIMESTAMP NULL

Foreign Keys
category_id → categories.id

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7. PRODUCT_VARIANTS
Purpose:
Size / Weight Variations
Example:
Tomato

- **500g**
- **1kg**
- **2kg**

Fields

Field	Type
id	UUID PK
product_id	UUID FK
sku	VARCHAR(100) UNIQUE
variant_name	VARCHAR(100)
unit	VARCHAR(50)

Field	Type
price	NUMERIC(10,2)
mrp	NUMERIC(10,2)
is_active	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP

Foreign Keys

product_id → products.id

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8. INVENTORY

Purpose:

Stock Management

Fields

Field	Type
id	UUID PK
variant_id	UUID FK
available_stock	INTEGER
reserved_stock	INTEGER
low_stock_threshold	INTEGER
created_at	TIMESTAMP
updated_at	TIMESTAMP

Foreign Keys

variant_id → product_variants.id

Formula:

Sellable Stock

=

available_stock - reserved_stock

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9. ORDERS

Purpose:

Customer Orders

Fields

Field	Type
id	UUID PK
user_id	UUID FK
address_id	UUID FK
order_number	VARCHAR(50) UNIQUE
subtotal	NUMERIC(10,2)
delivery_fee	NUMERIC(10,2)
total_amount	NUMERIC(10,2)
current_status	VARCHAR(50)
payment_status	VARCHAR(50)

Field	Type
notes	TEXT
created_at	TIMESTAMP
updated_at	TIMESTAMP

Foreign Keys

user_id → users.id

address_id → addresses.id

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10. ORDER_ITEMS

Purpose:

Products inside orders

Fields

Field	Type
id	UUID PK
order_id	UUID FK
variant_id	UUID FK
product_name_snapshot	VARCHAR(255)
unit_price	NUMERIC(10,2)
quantity	INTEGER
total_price	NUMERIC(10,2)

Foreign Keys

order_id → orders.id

variant_id → product_variants.id

Snapshot fields prevent history loss.

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11. PAYMENTS

Purpose:

Payment Records

Fields

Field	Type
id	UUID PK
order_id	UUID FK
payment_method	VARCHAR(50)
transaction_reference	VARCHAR(255)
payment_proof_url	TEXT
amount	NUMERIC(10,2)
payment_status	VARCHAR(50)
verified_by	UUID NULL
verified_at	TIMESTAMP NULL
created_at	TIMESTAMP

Foreign Keys

order_id → orders.id

verified_by → admins.id

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12. ADMINS
Purpose:
Back Office Users
Fields

Field	Type
id	UUID PK
full_name	VARCHAR(150)
email	VARCHAR(255) UNIQUE
password_hash	TEXT
role	VARCHAR(50)
is_active	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP

Roles:
SUPER_ADMIN

MANAGER

STAFF

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13. ORDER_STATUS_HISTORY
Purpose:
Complete Order Timeline
Fields

Field	Type
id	UUID PK
order_id	UUID FK
status	VARCHAR(50)
remarks	TEXT
updated_by_admin	UUID NULL
created_at	TIMESTAMP

Foreign Keys
order_id → orders.id

updated_by_admin → admins.id
Statuses:
PLACED

CONFIRMED

PACKED

OUT_FOR_DELIVERY

DELIVERED

CANCELLED

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14. FUTURE TABLES
Vendors
vendors

vendor_products
Supports marketplace model.

Notifications
notifications
notification_logs
WhatsApp
SMS
Email
Push

Coupons
coupons

coupon_usages

Reviews
reviews
Product reviews and ratings.

Loyalty Program
loyalty_accounts

loyalty_transactions
Reward points system.

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15. Recommended Indexes
users.mobile

products.slug

categories.slug

orders.order_number

orders.user_id

orders.current_status

order_items.order_id

inventory.variant_id

payments.order_id
Composite Indexes
(product_id, is_active)

(user_id, created_at)

(order_id, status)

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16. SQL CREATE TABLE (Core Schema)
Enable UUID
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

USERS

```
CREATE TABLE users (
   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
   full_name VARCHAR(150) NOT NULL,
   mobile VARCHAR(15) UNIQUE NOT NULL,
   email VARCHAR(255),
   password_hash TEXT NOT NULL,
   is_active BOOLEAN DEFAULT TRUE,
   created_at TIMESTAMP DEFAULT NOW(),
   updated_at TIMESTAMP DEFAULT NOW(),
   deleted_at TIMESTAMP
);
```

CATEGORIES

```
CREATE TABLE categories (
   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
   name VARCHAR(100) NOT NULL,
   slug VARCHAR(120) UNIQUE NOT NULL,
   description TEXT,
   image_url TEXT,
   is_active BOOLEAN DEFAULT TRUE,
   created_at TIMESTAMP DEFAULT NOW(),
   updated_at TIMESTAMP DEFAULT NOW()
);
```

PRODUCTS

```
CREATE TABLE products (
   id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
   category_id UUID NOT NULL REFERENCES categories(id),
   name VARCHAR(255) NOT NULL,
   slug VARCHAR(255) UNIQUE NOT NULL,
   description TEXT,
   image_url TEXT,
   is_active BOOLEAN DEFAULT TRUE,
```

```
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW(),
  deleted_at TIMESTAMP
);
```

PRODUCT_VARIANTS

```
CREATE TABLE product_variants (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  product_id UUID NOT NULL REFERENCES products(id),
  sku VARCHAR(100) UNIQUE NOT NULL,
  variant_name VARCHAR(100),
  unit VARCHAR(50),
  price NUMERIC(10,2) NOT NULL,
  mrp NUMERIC(10,2),
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

INVENTORY

```
CREATE TABLE inventory (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  variant_id UUID UNIQUE NOT NULL REFERENCES product_variants(id),
  available_stock INTEGER DEFAULT 0,
  reserved_stock INTEGER DEFAULT 0,
  low_stock_threshold INTEGER DEFAULT 5,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

(Continue remaining tables similarly during implementation phase.)